

Request for Budgetary Technical and Commercial Proposal No. OVE75-RFQ-08

Subject: Charge preparation: ZC feeders, 4488 DN-U belt dosers, SSh-25 screw mixer, UKV belt conveyors.

KVANT LLC (Russian Federation), within the Basic Engineering (FEED) stage of the construction project of a copper production plant — a roasting-leaching-electrowinning complex with a capacity of 75,000 tonnes of copper cathodes per year — kindly requests your budgetary technical and commercial proposal (budgetary quotation) for the item above. Project site: an operating metallurgical site in north-west Russia.

Brief description of the item: Feeders ZC-300/800/1200/1500; belt dosers up to 25 t/h; mixer 10–25 t/h, 37 kW, working volume 3.8 m³. Technical requirements are given in Annex 1 to this request.

Please include in your quotation:

1. budget price, quoted separately on EXW (manufacturer's works) and DDP project site in north-west Russia terms (Incoterms 2020), stating the currency and the validity period of the offer;
2. manufacturing lead time and estimated delivery time (in weeks from the date of order / advance payment);
3. overall dimensions and weights of one equipment unit and of the largest shipping blocks (for logistics assessment);
4. reference list of similar supplies: product, process medium and parameters, year of supply, country;
5. materials of parts in contact with the process medium, and the scope of supply (instrumentation, spare parts, documentation);
6. utility consumption and installation requirements (if available).

This is a budgetary enquiry at the Basic Engineering (FEED) stage: it does not constitute an offer and creates no obligations for either party; detailed technical data sheets will be issued at the next project stage.

Reply deadline: we kindly ask you to submit your quotation within 3 (three) weeks from receipt of this request to stepan@kvantpro.com. Please confirm receipt of this request.

Confidentiality. This request and the attached technical materials are provided solely for the preparation of the quotation, are confidential, and shall not be disclosed to third parties, copied or published without the prior written consent of KVANT LLC. Submission of a quotation creates no legal obligations for either party.

Contact person: Stepan Kharkov, KVANT LLC, stepan@kvantpro.com.

Annex 1: Technical requirements for the requested item.

Annex 1. Technical Requirements

To RFQ No. OVE75-RFQ-08. Parameters are based on the Customer's input data and design assignment; Basic Engineering (FEED) stage — values may be refined at the next project stage.

RFQ item	Charge preparation: ZC feeders, 4488 DN-U belt dosers, SSh-25 screw mixer, UKV belt conveyors
Summary	Feeders ZC-300/800/1200/1500; belt dosers up to 25 t/h; mixer 10–25 t/h, 37 kW, working volume 3.8 m ³
Specific requirements	Stainless steel execution on cement copper lines and on lines handling charge containing cement copper

Equipment covered by this request:

1. Belt conveyors (incl. reversible) Charge preparation

Type / model	UKV-320-Ts-60 / UKV-200-Ts-60 / UKV-200-Ts-37.5 / UKV-320-Ts-95
Duty	Transport of copper concentrate, copper-bearing cake, recycled materials and roasting products
Key parameters	Capacity up to 15 t/h
Quantity	To be confirmed at the next design stage; 5 reversible units for loading of the charge bins and the fluidized-bed (FB) furnace
Materials	Carbon steel, rubber
Dimensions, hold-up	Belt width 400 mm, average length 3000 mm, layer height 100 mm
Power consumption	3.4–30 kW

2. Belt conveyors for cement copper Charge preparation

Duty	Transport of cement copper
Key parameters	Capacity up to 15 t/h
Quantity	To be confirmed at the next design stage
Materials	Stainless steel, acid-resistant rubber

3. Screw conveyors Charge preparation

Duty	Transport of dusty high-temperature charge components (except cement copper) and roasting products
Key parameters	Capacity up to 15 t/h
Quantity	2 for charge feeding; separate units for waste-heat boiler, cyclone and electrostatic precipitator dust
Materials	Carbon steel
Dimensions, hold-up	Ø320 L10000 and L1000 mm; for waste-heat boiler dust Ø360 L5000; cyclone dust Ø360 L5000; electrostatic precipitator dust Ø200 L5000

4. Screw conveyors for cement copper Charge preparation

Duty	Transport of cement copper and of charge containing it
Key parameters	Capacity up to 15 t/h

Quantity	To be confirmed
Materials	Stainless steel

5. Receiving bins Charge preparation

Type / model	Cylindrical
Duty	Storage of a shift stock of copper concentrate, copper-bearing cake and cement copper, of calcine for FB furnace start-up, and of returned electrostatic precipitator dust
Key parameters	Volume sufficient for continuous operation
Quantity	Concentrate 60 m ³ , cake 20 m ³ , cement copper 10 m ³ , recycled materials 50 m ³
Materials	Carbon steel, PE500 polyethylene lining

6. Charge day bins Charge preparation

Type / model	Cylindrical
Duty	Uninterrupted feeding of the FB furnace with charge of constant composition
Key parameters	Volume per shift — 60 m ³
Quantity	2
Materials	Carbon steel, PE500 polyethylene lining

7. Disc (plate) feeders Charge preparation

Type / model	ZC-300 / ZC-800 / ZC-1200 / ZC-1500
Duty	Metered feeding of charge components to mixing and feeding of charge to the FB furnace
Key parameters	ZC-300 up to 3.6 t/h; ZC-800 up to 8 t/h; ZC-1200 up to 18 t/h; ZC-1500 up to 22 t/h
Quantity	2×ZC-300, 1×ZC-800, 1×ZC-1200, 2×ZC-1500
Materials	ZC-300 (cement copper) and ZC-1500 (charge) — stainless steel; others — carbon steel
Dimensions, hold-up	Concentrate Ø1200 H100; cake Ø800 H100; copper Ø300 H100; recycled materials Ø300 H100 mm
Power consumption	0.75–3 kW

8. Belt dosers Charge preparation

Type / model	4488 DN-U-1-X-40-1800-0-0.X (X = 3.2; 4; 5; 16; 25)
Duty	Control of the feed of charge components to mixing and of charge to the FB furnace
Key parameters	Per index X: up to 3.2 / 4 / 5 / 16 / 25 t/h
Quantity	1 unit each for X = 3.2; 4; 5; 16, and 2 units for X = 25
Materials	X = 3.2 and 25 — stainless steel and acid-resistant rubber; others — carbon steel and standard rubber
Dimensions, hold-up	Belt 500 mm, linear load 40 kg/m

Power consumption	0.55–1.5 kW
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9. Screw mixer Charge preparation

Type / model	SSh-25
Duty	Preparation of charge for roasting in the FB furnace
Key parameters	10–25 t/h
Quantity	1
Materials	Stainless steel
Dimensions, hold-up	Working volume 3.8 m ³
Power consumption	37 kW

The quotation must confirm: process medium, temperature, pressure, materials of wetted parts, capacity, dimensions and weight, power consumption, manufacturing lead time, delivery terms.

预算性技术商务报价征询函 编号 OVE75-RFQ-08

事由：炉料制备：ZC 给料机、4488 DN-U 皮带定量给料机、SSh-25 螺旋混合机、UKV 皮带输送机。

KVANT LLC (俄罗斯联邦) 正在开展铜冶炼厂建设项目 (焙烧—浸出—电积联合装置，年产阴极铜 75,000 t) 的基础设计 (FEED)，特请贵方就上述设备提交预算性技术商务报价 (预算报价)。项目场地：俄罗斯西北部一处运营的冶金厂区。

设备简述：给料机 ZC-300/800/1200/1500；皮带定量给料机最大 25 t/h；混合机 10-25 t/h、37 kW、有效容积 3.8 m³。技术要求见本函附件 1。

请在报价中包含以下内容：

1. 预算价格——分别按 EXW (制造厂交货) 和 DDP 项目场地 (俄罗斯西北部) 条件 (Incoterms 2020) 报价，注明币种及报价有效期；
2. 制造周期及预计交货期 (自订单/预付款之日起按周计)；
3. 外形尺寸及重量：单台设备及最大运输单元的尺寸和重量 (用于物流分析)；
4. 类似供货业绩：产品、工艺介质及参数、供货年份、国家；
5. 与工艺介质接触部件的材质及供货范围 (仪表、备件、技术文件)；
6. 能源介质消耗及安装要求 (如有资料)。

本询价为预算性询价，属于基础设计 (FEED) 阶段：不构成要约，对双方不产生任何义务；详细询价数据表将在项目下一阶段发出。

回复期限：请自收到本函之日起 3 (三) 周内将报价发送至 stepan@kvantpro.com。请确认收到本函。

保密条款。本询价函及所附技术资料仅供编制报价使用，属保密信息，未经 KVANT LLC 事先书面同意，不得向第三方披露、复制或发布。提交报价不对双方构成法律义务。

联系人：Stepan Kharkov, KVANT LLC, stepan@kvantpro.com。

附件 1：询价设备技术要求。

附件 1. 技术要求

对应询价函编号 OVE75-RFQ-08，参数依据业主输入数据及设计任务书（基础设计/FEED 阶段），后续阶段可能进一步细化。

询价项目	炉料制备：ZC 给料机、4488 DN-U 皮带定量给料机、SSh-25 螺旋混合机、UKV 皮带输送机
简述	给料机 ZC-300/800/1200/1500；皮带定量给料机最大 25 t/h；混合机 10-25 t/h、37 kW、有效容积 3.8 m³
特殊要求	置换铜物料线及含置换铜炉料线的设备采用不锈钢材质

本询价涵盖的设备清单：

1. 皮带输送机（含可逆式） 炉料制备

型式/型号	UKV-320-Ts-60 / UKV-200-Ts-60 / UKV-200-Ts-37.5 / UKV-320-Ts-95
用途	输送铜精矿、含铜滤饼、返回料及焙烧产物
主要参数	处理能力最大 15 t/h
数量	待下一设计阶段确定；其中 5 台可逆式用于炉料仓及沸腾焙烧炉上料
材质	碳钢、橡胶
尺寸、持料量	带宽 400 mm，平均长度 3000 mm，料层高度 100 mm
电耗	3.4-30 kW

2. 置换铜皮带输送机 炉料制备

用途	输送置换铜
主要参数	处理能力最大 15 t/h
数量	待下一设计阶段确定
材质	不锈钢、耐酸橡胶

3. 螺旋输送机 炉料制备

用途	输送高温粉尘性炉料组分（置换铜除外）及焙烧产物
主要参数	处理能力最大 15 t/h
数量	2 台用于炉料给料；余热锅炉、旋风除尘器、静电除尘器粉尘各设专用输送机
材质	碳钢
尺寸、持料量	Ø320 L10000 及 L1000 mm；余热锅炉粉尘 Ø360 L5000；旋风除尘器粉尘 Ø360 L5000；静电除尘器粉尘 Ø200 L5000

4. 置换铜螺旋输送机 炉料制备

用途	输送置换铜及含置换铜的炉料
主要参数	处理能力最大 15 t/h
数量	待定

材质	不锈钢
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5. 受料仓 炉料制备

型式/型号	圆筒形
用途	储存铜精矿、含铜滤饼、置换铜的班储量及沸腾炉开炉用焙砂，接收静电除尘器返回粉尘
主要参数	容积满足连续运行要求
数量	精矿 60 m ³ ，滤饼 20 m ³ ，置换铜 10 m ³ ，返回料 50 m ³
材质	碳钢，内衬 PE500 聚乙烯

6. 炉料供料仓 炉料制备

型式/型号	圆筒形
用途	向沸腾焙烧炉不间断供给成分稳定的炉料
主要参数	每班容积——60 m ³
数量	2
材质	碳钢，内衬 PE500 聚乙烯

7. 圆盘给料机 炉料制备

型式/型号	ZC-300 / ZC-800 / ZC-1200 / ZC-1500
用途	炉料组分定量给入混合工序，并将炉料给入沸腾焙烧炉
主要参数	ZC-300 最大 3.6 t/h；ZC-800 最大 8 t/h；ZC-1200 最大 18 t/h；ZC-1500 最大 22 t/h
数量	2×ZC-300、1×ZC-800、1×ZC-1200、2×ZC-1500
材质	ZC-300（置换铜）及 ZC-1500（炉料）为不锈钢；其余为碳钢
尺寸、持料量	精矿 Ø1200 H100；滤饼 Ø800 H100；置换铜 Ø300 H100；返回料 Ø300 H100 mm
电耗	0.75-3 kW

8. 皮带定量给料机 炉料制备

型式/型号	4488 DN-U-1-X-40-1800-0-0.X (X = 3.2 ; 4 ; 5 ; 16 ; 25)
用途	调节炉料组分给入混合工序及炉料给入沸腾焙烧炉的流量
主要参数	按指标 X：最大 3.2 / 4 / 5 / 16 / 25 t/h
数量	X = 3.2、4、5、16 各 1 台；X = 25 为 2 台
材质	X = 3.2 及 25 为不锈钢加耐酸橡胶；其余为碳钢加普通橡胶
尺寸、持料量	带宽 500 mm，线载荷 40 kg/m
电耗	0.55-1.5 kW

9. 螺旋混合机 炉料制备

型式/型号	SSh-25
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用途	制备沸腾焙烧炉用炉料
主要参数	10-25 t/h
数量	1
材质	不锈钢
尺寸、持料量	有效容积 3.8 m ³
电耗	37 kW

报价中须确认：工艺介质、温度、压力、接触部件材质、处理能力、尺寸与重量、电耗、制造周期、交货条件。